

SymPy Bug Card

Bug 10: principal cube-root inequality includes negatives

Task. Solve the inequality over real x as represented in SymPy by `x**(S(1)/3) <= 1`.

$$x^{1/3} \leq 1$$

SymPy's answer.

$$(-\infty, 1].$$

Correct answer.

$$[0, 1].$$

Explanation. The expression `x**(S(1)/3)` is a principal `Pow`, not SymPy's real-root operation. Thus, unlike the real cube root, SymPy's principal value at $x = -1$ is

$$(-1)^{1/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$$

so the ordered real comparison with 1 is not satisfied there. On the real-valued domain $x \geq 0$, the principal cube root is monotone and $x^{1/3} \leq 1$ gives $0 \leq x \leq 1$.

Likely root cause. The diagnosis points to `continuous_domain` treating odd-denominator fractional powers as real-valued over all real bases, which matches `real_root` semantics rather than principal `Pow` semantics; the inequality solver then samples and accepts an interval containing negative inputs. Deduplication frames this as a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-010-principal-cube-root-inequality-includes-negatives/](#)